

ASHLAND JFK, WI, USA

WMO: 726419

Lat: 46.549N

Lon: 90.919W

Elev: 826

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 94929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-7.9	-20.0	1.9	-11.1	-16.1	2.4	-7.1	23.2	17.9	20.6	17.6	8.4	240	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	86.5	71.5	83.7	69.8	81.2	68.0	74.3	83.3	72.2	80.6	70.2	78.0	10.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.5	119.9	79.4	69.4	111.3	77.6	67.1	102.9	75.1	38.3	83.2	36.4	80.8	34.6	77.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.8	16.8	-18.2	92.3	4.6	3.1	-21.4	94.6	-24.1	96.4	-26.6	98.1	-30.0	100.4
			-18.7	77.6	4.3	1.9	-21.8	79.0	-24.3	80.1	-26.7	81.2	-29.8	82.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	14.6	16.9	28.0	39.7	51.1	60.8	67.5	65.4	58.2	45.1	33.1	20.2	
	DBStd	20.65	12.48	11.57	12.23	9.07	8.59	7.15	6.27	5.89	8.24	8.53	10.02	11.28	
	HDD50	4794	1098	926	687	329	89	4	0	0	19	205	513	924	
	HDD65	8736	1563	1346	1146	758	443	165	47	68	234	620	957	1389	
	CDD50	1818	0	0	6	20	122	328	542	476	265	53	6	0	
	CDD65	286	0	0	1	0	10	38	125	79	30	3	0	0	
	CDH74	3097	0	0	5	22	172	476	1323	769	301	29	0	0	
	CDH80	805	0	0	0	4	48	115	392	177	66	3	0	0	
Wind	WSAvg	7.5	8.2	8.4	7.8	8.5	7.6	6.4	6.3	6.2	6.9	7.5	7.9	8.0	
Precipitation	PrecAvg	30.4	1.2	0.9	1.7	2.6	3.2	3.4	3.9	4.0	3.7	2.5	2.2	1.3	
	PrecMax	41.4	4.4	2.9	5.9	9.1	7.5	6.6	9.0	8.8	10.0	5.9	8.5	4.6	
	PrecMin	20.1	0.0	0.0	0.2	0.6	0.6	1.4	0.6	1.3	0.9	0.4	0.3	0.1	
	PrecStd	5.2	1.0	0.7	1.1	1.6	1.7	1.4	2.0	1.7	1.9	1.5	1.7	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	49.4	68.6	77.0	86.1	88.2	91.8	88.4	86.3	78.6	64.0	45.7	
		MCWB	37.4	44.3	56.3	59.0	65.5	72.5	74.6	72.9	72.1	62.5	52.6	39.5	
	2%	DB	38.9	42.0	57.1	68.3	78.9	83.2	87.4	84.3	81.5	71.3	58.1	40.9	
		MCWB	35.0	36.2	46.4	52.1	61.4	68.8	72.8	70.8	68.3	59.4	50.7	37.4	
	5%	DB	34.8	36.6	50.1	62.0	72.6	79.5	84.1	81.5	77.4	64.5	52.8	37.4	
		MCWB	32.5	32.5	43.0	49.4	58.2	66.0	70.2	69.2	67.0	55.8	45.8	34.9	
	10%	DB	31.8	33.1	44.4	56.2	67.6	75.5	81.5	78.6	73.3	59.4	47.9	35.0	
		MCWB	29.7	30.2	38.1	46.0	56.2	64.3	68.8	67.2	64.6	51.6	42.3	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	42.2	59.9	60.4	69.7	75.0	77.7	75.7	74.7	65.5	56.6	41.5
			MCDB	42.9	48.6	67.1	72.8	79.6	85.2	87.6	84.1	82.5	74.9	61.7	43.1
2%		WB	35.2	36.6	48.8	54.2	64.7	71.3	74.7	73.3	71.0	60.2	51.2	37.8	
		MCDB	38.2	41.1	56.4	64.7	73.9	80.4	84.1	81.9	78.6	68.0	57.3	40.3	
5%		WB	32.8	33.0	43.0	50.3	60.8	68.3	72.6	71.1	68.2	56.7	46.2	35.4	
		MCDB	35.0	36.2	48.8	61.1	70.4	76.3	81.0	78.7	75.1	63.9	51.0	37.3	
10%		WB	29.6	30.2	38.8	46.5	57.3	65.6	70.5	68.9	65.6	53.0	43.0	33.1	
		MCDB	31.6	32.7	43.7	55.3	65.6	73.0	78.5	76.1	72.5	58.1	47.8	34.9	
Mean Daily Temperature Range	5% DB	MDBR	15.1	17.4	18.8	20.4	23.5	23.0	23.3	22.3	21.6	18.2	14.9	13.2	
		MCDBR	15.6	19.8	24.8	32.9	32.1	27.7	26.0	25.7	26.0	26.6	23.1	15.2	
		MCWBR	12.7	14.8	16.2	19.3	17.5	15.3	13.1	13.8	14.9	16.3	15.8	12.3	
	5% WB	MCDBR	15.0	17.5	22.7	30.5	28.9	25.3	23.8	22.9	22.5	23.2	21.7	13.9	
		MCWBR	12.5	13.5	15.9	18.6	17.2	15.6	13.4	13.5	13.9	15.5	15.8	11.7	
Clear-Sky Solar Irradiance	taub	0.250	0.281	0.308	0.345	0.374	0.373	0.375	0.376	0.359	0.315	0.290	0.266		
	taud	2.441	2.353	2.388	2.394	2.365	2.393	2.421	2.429	2.456	2.557	2.524	2.454		
	Ebn at Noon	277	285	292	289	283	283	280	276	272	271	258	255		
	Edh at Noon	21	28	32	35	37	37	35	34	30	23	20	19		
All-Sky Solar Radiation	RadAvg	421	755	1133	1427	1687	1832	1926	1607	1215	738	428	297		
	RadStd	43	72	100	148	124	126	75	104	70	88	54	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page